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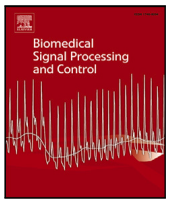

# MGFormer: A lightweight multi-granular transformer for subject-independent Alzheimer's classification

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ABSTRACT

Transformer-based models have shown great potential for detecting Alzheimer's Disease (AD) from EEG signals. However, these models often struggle due to their high computational demands and limited ability to capture EEG features across multiple scales. Most existing methods use shallow feature extraction techniques, failing to fully consider important spatial relationships and spectral information essential for identifying subtle disease-related changes. We propose MGFormer, a lightweight Convolutional Neural Network (CNN)-Transformer hybrid architecture designed for Electroencephalogram (EEG)-based AD classification. MGFormer features a Multi-Granular Token Encoder (MgTE) that extracts spatial–temporal features at multiple granularities, and a Hybrid Feature Fusion (HFF) module that combines long-range temporal modeling (via self-attention), local feature extraction (via 1D CNN), and Fast Fourier based spectral information from the embeddings. We perform a subject-independent evaluation across 113 subjects, which ensures realistic generalizability in medical domain. MGFormer achieves 70.48% accuracy on the AUT-AD dataset and 97.85% on the FSU-AD dataset, outperforming nine state-of-the-art models across six metrics. Ablation studies confirm the critical contributions of FFT features, 1D CNN, and HFF depth, while model complexity analysis shows superior efficiency in parameters and FLOPs compared to baselines. MGFormer effectively addresses key limitations in EEG-based AD classification, offering strong accuracy, generalizability, and computational efficiency.

## 1. Introduction

Alzheimer's Disease (AD) is the most common subtype of progressive dementia worldwide. Its diagnosis typically relies on caregivers' observations, often resulting in delayed identification [1]. Early diagnosis of AD reflects a high-quality healthcare sector, significantly aiding patients by slowing cognitive decline and potentially reversing memory loss [2]. However, current diagnostic methods for AD are predominantly invasive, including blood tests, genetic screening, and cerebrospinal fluid (CSF) analysis [3]. These invasive techniques are expensive and need expert interpretation. Within the last several decades, electroencephalogram (EEG) signals have emerged as a promising, non-invasive, and cost-effective alternative [4]. Resting-state EEG data, in particular, have gained attention for their ability to characterize AD due to their accessibility and relatively high signal-to-noise ratio [5].

EEG signals are recorded using multiple electrodes placed on the scalp, capturing electrical potentials between electrodes and the brain [4]. Each electrode, known as a channel, outputs a temporal signal, creating time-series data characterized by both temporal and spatial dimensions in the shape $(channels, time\ steps)$. Initial EEG studies on AD detection employed statistical measures like mean, median, skewness, wavelet transforms and geometric features [6,7]. Similarly, supervised network-based fuzzy learning models demonstrated promising results in AD identification by integrating EEG-derived network features with fuzzy systems, significantly boosting classification accuracy [8]. In parallel, research on acupuncture's effects on brain function has adopted similar methodologies, such as employing EEG signals to explore the modulation effects of acupuncture and to classify different acupuncture manipulations using machine learning and graph theory [9]. These studies reveal that acupuncture not only alters spectral power and synchronization patterns but also reconfigures functional brain networks, emphasizing its potential impact on the central nervous system [9,10]. Graph signal processing was also widely used because of its brain-connectivity features [11]. Traditional machine learning approaches typically extracted these handcrafted features from EEG data to train predictive models [12]. However, recent literature suggests that these traditional methods inadequately capture critical spatial–temporal features inherent in EEG signals [13,14]. To address these shortcomings, convolutional neural networks (CNNs) have been proposed [15–18],

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seizure detection, often surpassing traditional feature-based classifiers in accuracy [33]. Some studies transform EEG data into images to leverage the power of 2D CNNs for AD analysis. This includes using power spectrum density (PSD) across channels and extracting spectrum energy features [14,31]. Another method involves transforming EEG sub-bands (theta, alpha, and beta) into 2D images or converting EEG signals into time-frequency (TF) images using the short-time Fourier transform (STFT) for training with 2D-CNNs [34]. Another advantage of using CNN is they are extremely good at feature extraction without extensive pre-processing [35]. However, CNN's with small kernel size often ignores long-term temporal information [14]. Similarly, CNN assumed locality, where adjacent data points are supposedly more relevant than distant one's [28], which might be problematic as EEG signals may show patterns over a distant range. As a result, inter-subject variability results in subject-independent setups showcases way lower accuracy than subject-dependent setup [25].

### 2.2. Transformer for EEG based disease classification

Since its inception, transformers [24] have been used in a wide range of sequential tasks, namely in the field of Large Language Models (LLM). Following time-series data and its sequential nature, several studies have introduced the idea of using transformers within time-series domain [36,37]. For medical time-series data such as EEG and ECG, transformers performs superior in many tasks because of its inherent capability of catering to long-range dependencies [14]. An example of it is the adaptation of Vision Transformer (ViT) into EEG domain, which involves partitioning the data into non-overlapping segments along the temporal dimension and converting them into tokens using a linear projection layer [27]. This allows the Transformer to process EEG effectively, similar to how it processes sequences of image patches. Models like iTransformer [26] apply attention on inverted dimensions (variates) to capture multivariate correlations, which was be adapted for EEG as well. The attention mechanism allows the model to dynamically adjust its focus, applying weighted attention to critical signal features within the EEG data, such as specific frequency band rhythm changes. This has significantly enhance the CTNet's sensitivity to key information and improve decoding accuracy [38]. When combined with CNNs, hybrid architectures such as Deformer [14] and ADFormer [13] leverages the strengths of both models: CNNs for effective local feature extraction and Transformers for capturing global dependencies and long-range context, leading to improved performance compared to using either architecture alone. The combination of CNN and Transformer architectures often yields better performance than using either alone [13]. However, transformer's have an inherent problem with quadratic complexity, which refers to the increasing time and space complexity with sequence length [24]. In the case of hybrid CNN-transformer architecture, this boils down to having the right CNN layers for optimal kernel selection and finding a suitable transformer architecture. In Deformer [14], the kernel size was chosen according to the frequency of EEG signal, with aggressive max pooling to reduce number of tokens. Similarly, [39] has used average pooling for the same reason mentioned. However, ADFormer [13] represents each individual time-series data point as a separate token, potentially leading to increased training and inference time.

## 3. Method

We present an extensive description of MGFormer's architecture, explicitly tailored for AD detection and inspired by principles of multi-granular convolutions, transformer-based attention, and frequency-domain feature extraction. This section describes each component, highlighting the granularities and the design choices that allow the model to capture relevant features.

### 3.1. Preliminaries

Let $\mathbf{X} \in \mathbb{R}^{B\times C\times T}$ represent a batch of EEG segments, where: $B$ is the batch size (number of samples per batch), $C$ is the number of EEG channels, and $T$ is the number of time points in each segment (e.g., $T = 256$ for one second of EEG sampled at 256 Hz). Now, let $f$ be the EEG sampling frequency (e.g., $f = 256\,\text{Hz}$). We define a set of temporal granularities $\mathcal{G} = 0.04, 0.06, 0.08$ (in seconds). The 0.04 value (40 ms) is chosen to capture rapid neural transients, such as microstates. The 0.06 value (60 ms) reflects slightly slower, short-lived transient features as well as emerging oscillatory activity. Lastly, 0.08 (80 ms) emphasizes slower oscillations and their modulations [40]. As described below, each value $x \in \mathcal{G}$ is used to compute a fixed kernel size $\kappa$ via $\kappa = \lfloor f \times x \rfloor$, where $f$ is frequency of the signal. In essence, the model's convolutional filters and tokenization steps incorporate multiple $\kappa$ values to capture multi-scale granular features in the EEG signal.

### 3.2. Multi-granular token encoding (MgTE)

From a medical standpoint, the brain operates as an integrated system, leading to the intuitive assumption that inherent correlations exist among different channels [13,32], as each channel reflects the activity of distinct brain or heart regions. Intuitively, a method that retains both temporal and spatial correlations can more effectively capture the dynamic interactions between these regions, leading to a better and accurate representation of physiological signals.

To effectively capture both temporal and spatial features, we employ the Multi-granular Token Encoding (MgTE) module, which operates at three different granularities in $\mathcal{G}$. As illustrated in Fig. 2, MgTE consists of two sub-modules: the first focuses on temporal feature extraction, while the second encodes spatial information. In the temporal encoding module, we utilize $\mathcal{G}$ to define kernel sizes, where each kernel size $P_i \in \mathcal{G}$ segments the input $\mathbf{X}$ along the temporal dimension into contiguous, non-overlapping patches. Each segment is represented as $\mathbb{R}^{C\times l_i}$. The extracted features are then concatenated, followed by max-pooling for feature reduction and batch normalization for stable learning. Once temporal convolution is applied, the spatial encoding module (channel-wise kernel) integrates information across the $C$ channels to capture spatial dependencies. Specifically, we implement channel-wise convolution, which operates across the feature map's channel dimensions-similar to how Sception spans the entire electrode set in [29]. This step is followed by batch normalization and max-pooling to ensure stable training and minimize the number of tokens. Finally, positional encoding is added to the tokens to preserve order information. By using multi-scale tokenization, our tokenizer captures both high-frequency local fluctuations (short patches) and broader contextual information (long patches) [14,29].

Since transformers are permutation invariant [24], and they do not rely on recurrence or convolution for explicit ordering, we add a positional encoding (PE) to preserve the temporal ordering of EEG signals. For a position $t \in [0, \ldots, N-1]$ and feature index $d \in [0, \ldots, D-1]$, we define:

Even indices $(2i)$:

$$\text{PE}(t, 2i) = \sin\left(t \cdot e^{-\ln(10000)\,(2i/D)}\right), \tag{1}$$

Odd indices $(2i+1)$:

$$\text{PE}(t, 2i+1) = \cos\left(t \cdot e^{-\ln(10000)\,(2i/D)}\right). \tag{2}$$

We create a matrix $\mathbf{P} \in \mathbb{R}^{N\times D}$ of these values and add it elementwise to the token embeddings. This step ensures the model is aware of time-step $t$ within each segment. This enriched feature representation is then directly fed into the attention and fusion blocks described in the following section.

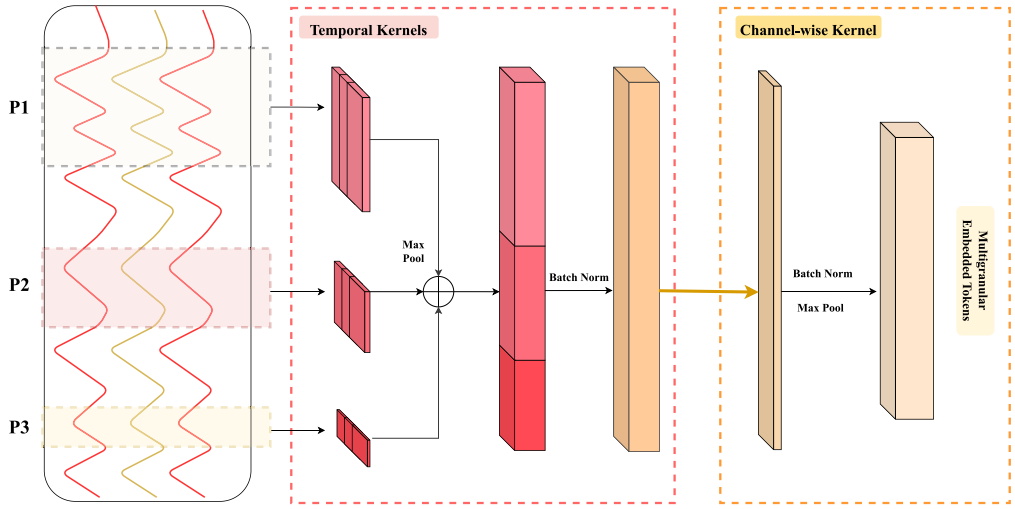



**Fig. 2. Illustration of the Multi-granular Token Encoding (MgTE) module.** MgTE extracts and encodes EEG features at multiple granularities $\kappa \in \{P_1, P_2, P_3\}$, preserving temporal dynamics through multi-scale temporal convolution and spatial correlations through channel-wise convolution. The resulting tokens combine both local and global temporal and spatial information.

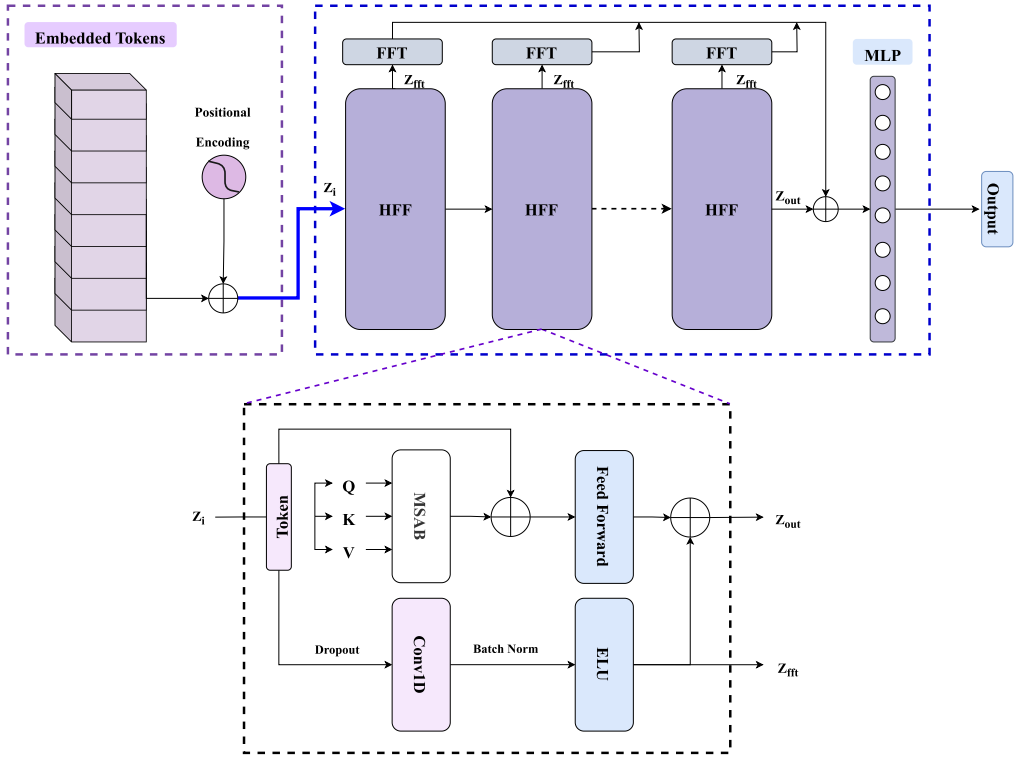



**Fig. 3. Detailed diagram of Hybrid Feature Fusion module.** Embedded tokens enriched with positional encoding are processed through sequential HFF layers, each designed to model both local and global dependencies. Within each HFF block, multi-head self-attention captures long-range interactions, while convolutional operations preserve fine-grained local features. Frequency-domain representations are extracted via FFT from each HFF output and integrated alongside the final token representation. A multi-layer perceptron (MLP) then produces the final output, enabling rich spatiotemporal and spectral feature fusion.

### 3.3. Hybrid feature fusion (HFF)

Inspired by the parallel hybrid transformer blocks of Deformer [14], the Hybrid feature fusion (HFF) module integrates vanilla transformer-based sequence modeling [24] with a parallel convolutional processing path. It consists of transformer encoder layers augmented with a local convolution path and enriched with Fast Fourier spectral features. HFF is depicted in Fig. 3, where the tokens previously generated by MgTE are fed into the HFF layers with positional encoding.

At each layer $l$, we have:

$$\mathbf{z}_i^{(l)} \in \mathbb{R}^{B \times D \times N}.$$

This design allows the model to capture long-range dependencies in the EEG (via self-attention) while also preserving local temporal structure (via temporal convolutions), and to utilize both time-domain and frequency-domain information. Formally, each hybrid encoder block includes a Multi-Head Self-Attention layer, a position-wise Feed-Forward Network (FFN), and a 1D convolution, all with appropriate residual

connections and normalization (analogous to a Transformer encoder layer extended with a convolution module).

Let $\mathbf{z}^{(l)} \in \mathbb{R}^{B\times D\times N}$ denote the tokens entering layer $l$. Each layer contains:

**Multi-Head Self-Attention (MHSA).** We compute query, key, and value projections for $h$ heads:

$$\mathbf{Q} = \mathbf{z}\,\mathbf{W}_Q, \quad \mathbf{K} = \mathbf{z}\,\mathbf{W}_K, \quad \mathbf{V} = \mathbf{z}\,\mathbf{W}_V,$$

then apply scaled dot-product attention:

$$\mathrm{Attn}(\mathbf{Q},\mathbf{K},\mathbf{V}) = \mathrm{Softmax}\left(\frac{\mathbf{Q}\,\mathbf{K}^{\top}}{\sqrt{d_h}}\right)\mathbf{V}. \tag{3}$$

We concatenate across heads, project back to dimension $D$, and add a skip connection to preserve features:

$$\mathbf{z}_{\mathrm{attn}} = \mathbf{z} + \mathrm{MHSA}(\mathbf{z}). \tag{4}$$

**Convolution branch.** In parallel with attention, we apply a 1D convolution to $\mathbf{z}^{(l)}$. This helps model local temporal structure (microstate-scale patterns [14]). Like attention, the convolution branch includes batch normalization, an ELU activation, and a skip connection.

**Spectral extraction.** We compute the (log of) the mean amplitude spectrum of the convolution output:

$$\mathbf{Z}_{\mathrm{fft}} = \log\Big(\mathrm{mean}\big(|\mathcal{F}(\mathbf{z}_{\mathrm{conv}})|\big) + \epsilon\Big) \tag{5}$$

collecting these frequency features at each layer to capture EEG spectral signatures relevant to AD.

**Feed-forward network (FFN).**

After attention, we apply a position-wise FFN with expansion (factor of 4 in hidden dimension) to refine features:

$$\mathbf{z}^{\mathrm{ff}} = \mathbf{W}_2\Big(\mathrm{GELU}\big(\mathbf{W}_1\,\mathrm{LayerNorm}(\mathbf{z}_{\mathrm{attn}})\big)\Big). \tag{6}$$

Another skip connection integrates the FFN output with the residual from convolution.

Altogether, each HFF layer fuses time-domain attention with local convolution and frequency-domain features. After passing through $L$ such layers, we ean-pool across tokens, concatenate all collected FFT-based features from each layer, concatenate FFT-based features with the pooled output. Hence, the final output has dimension $\mathbb{R}^{B\times[D\times(L+1)]}$, which then goes into a linear classifier for final classification.

## 4. Experimental setups

### 4.1. Baseline

We chose five transformer based model, iTransformer [26], AD-Former [13], Crossformer [41], MTST [42], EEGViT [27], two CNN-based model EEGNet [43], TSCeption [29], one vanilla Long-short term memory (LSTM) [44] model and one CNN-Transformer model Deformer [14] to demonstrate our models capability. Appendix contains all the detailed breakdown of these models and their respective architectures.

### 4.2. Datasets

We evaluate the performance of MGFormer using two datasets, Dataset-I was collected from Florida State University (FSU-AD) [45], and Dataset-II from Aristotle University of Thessaloniki [46] (AUT-AD). A detailed description of the datasets and preprocessing are as follows (see Table 1 for detail):

#### 4.2.1. AUT-AD

The dataset from Aristotle University of Thessaloniki [46] consists of EEG resting-state recordings with eyes closed from 65 participants, including individuals diagnosed with AD and healthy control (HC) subjects. The diagnostic criteria were based on Mini-Mental State Examination (MMSE) scores, with participants classified into two subject groups (A and C) according to their health status: Group A included individuals with MMSE scores $\geq 24$ (cognitively healthy), and Group C included individuals with MMSE scores $< 24$ (cognitively impaired), with all recordings conducted under eyes-closed conditions. The recordings were acquired using 19 scalp electrodes and a sampling rate of 500 Hz. We have used files from the "derivatives" folder, which contain preprocessed and denoised EEG recordings. The preprocessed EEG recordings were split for subject-independent analysis, and each trial was segmented into 5-second epochs, without overlapping. This resulted in 10,615 samples.

#### 4.2.2. FSU-AD

The Florida State University Alzheimer's Dataset (FSU-AD) is a publicly available EEG time series dataset [45]. The dataset includes four subject groups (A–D) based on health status and eye state, where Groups A and B consist of healthy controls and Groups C and D include probable Alzheimer's patients diagnosed using MMSE scores $< 24$ and the DSM-III-R criteria. DSM-III-R requires that cognitive deficits significantly impair social or occupational functioning and are not attributable to other medical or psychiatric conditions. Additionally DSM-III-R is a qualitative metric unlike MMSE, which has a numerical threshold. The dataset contains 19 EEG channels recorded at a sampling rate of 128 Hz. The majority of EEG trials last 8 s, while some extend to 10, 12, or 14 s, with timestamps ranging from 1,024 to 1,792. Each trial is segmented into 1-second samples with 50% overlap, resulting in 768 samples, each containing 128 timestamps.

#### 4.2.3. Pre-processing

For the AUT-AD dataset [46], the authors provided preprocessed data in the "derivatives" folder, which included several standard EEG preprocessing steps: band-pass filtering in the range of 0.5–45 Hz, re-referencing to the average of electrodes A1 and A2, and artifact correction using Artifact Subspace Reconstruction (ASR). Additionally, Independent Component Analysis (ICA) was applied, and components identified as ocular or myogenic artifacts were removed using the ICLabel plugin in EEGLAB. Each trial was segmented into 5-second epochs, without overlapping. This resulted in 10,615 samples. For the FSU-AD dataset, preprocessing followed the procedure described in [13], which involved segmenting the original EEG recordings into 1-second epochs with 50% overlap. These recordings were band-limited to the range of 1–30 Hz and sampled at 128 Hz.

### 4.3. Implementation details

Conventionally, most studies applying deep neural network architectures to EEG uses subject dependent method, where both train and test sets can contain EEG segments from the same individual. However, this causes data leakage and the model's usually overfits [25]. In this work, we implement subject independent setup to train, validate and test our model. This method involves partitioning all subjects and the corresponding samples into training and test sets according to a predefined ratio or particular subject identifiers. The importance of this setup lies in its ability to evaluate real-world applicability. In the realm of dementia diagnosis, it is essential to create a model that can precisely diagnose dementia in unseen people. To guarantee a fair assessment, labeled samples from these unseen patients are excluded from the training phase. The distinctive features and noise present in each subject's data limits the identification of generalizable trends across various subjects. So models trained in a subject-independent way

**Table 1**
**Summary of participant demographics for the AUT-AD and FSU-AD datasets used in this study.** Each group is defined by clinical condition (Healthy Control or AD), eyes state during EEG recording (Open or Closed), and gender distribution. Age is reported as mean ± standard deviation. Mini-Mental State Examination (MMSE) scores [47] are included to assess cognitive impairment severity.

| Dataset | Group | Condition | Eyes State | Gender (M/F) | Age (Mean ± SD) | MMSE (Mean ± SD) |
|---|---|---|---|---|---|---|
| AUT-AD | A | Healthy Control | Closed | 12/24 | 66 ± 7.9 years | 30.0 ± 0.0 |
| | C | AD | Closed | 18/11 | 68 ± 5.4 years | 17.8 ± 4.5 |
| FSU-AD | A | Healthy Control | Open | 12/12 | 72 ± 5.8 years | 28.7 ± 2.9 |
| | B | Healthy Control | Closed | 12/12 | 72 ± 5.8 years | 28.7 ± 2.9 |
| | C | AD | Open | 10/14 | 69 ± 7.2 years | 19.4 ± 1.0 |
| | D | AD | Closed | 10/14 | 69 ± 7.2 years | 19.4 ± 1.0 |

generally exhibit inferior performance relative to those trained in a subject-dependent setup [13].

We use 20/80 data splitting, where roughly 20% of the individuals for any given dataset are used for the test-set, while remaining are kept for training. In both FSU-AD and AUT-AD we randomly chose the participants. We employed six evaluation metric: accuracy (Eq. (7)), precision (Eq. (8)), F1-score (Eq. (10)), AUROC (Eq. (11)) and AUPRC (Eq. (12)). Batch size was set 16 for FSU-AD and 64 for AUD-AD. The learning rate was 0.0001 with Adam optimizer and dropout rate 0.2 for both datasets. The model with best validation accuracy was saved and used for the test set. Each model was run for 50 epochs, 5 times using different random seed ranging from 41 to 45. We then averaged the test accuracy on these five random seeds to report the metrics. All experiments were run on a single T4 GPU, which was available on Kaggle.

$$\text{Accuracy} = \frac{\sum_{i=1}^{N}(\hat{y}_i = y_i)}{N} \tag{7}$$

$$\text{Precision} = \frac{\sum_k w_k \cdot \frac{TP_k}{TP_k+FP_k}}{\sum_k w_k} \tag{8}$$

$$\text{Recall} = \frac{\sum_k w_k \cdot \frac{TP_k}{TP_k+FN_k}}{\sum_k w_k} \tag{9}$$

$$\text{F1-score} = \frac{\sum_k w_k \cdot \frac{2 \cdot TP_k}{2TP_k+FP_k+FN_k}}{\sum_k w_k} \tag{10}$$

$$\text{AUROC} = \int_{-\infty}^{\infty} \text{TPR}(t)\, d\text{FPR}(t) \tag{11}$$

$$\text{AUPRC} = \int_{0}^{1} \text{Precision}(r)\, d\text{Recall}(r) \tag{12}$$

Where, $N$ = Number of samples, $y_i$ = True label, $\hat{y}_i$ = Predicted label, $TP_k, FP_k, FN_k$ are the true positives, false positives, and false negatives, $w_k$ is the weight.

## 5. Results & analysis

### 5.1. Train results (Loss curve)

To evaluate the training stability and convergence of various models, we analyzed the training loss curves for the AUT-AD and FSU-AD datasets across 50 epochs × 5 seeds. Fig. 4 provide comparative insights into model behavior. For both datasets, MGFormer model consistently demonstrated faster and smoother convergence compared to others. EEGNet and MTST showed relatively slower convergence and higher final loss values, reflecting limited representational capacity. Models such as Crossformer and iTransformer also showed competitive performance, stabilizing early with low loss values, albeit with minor fluctuations. Overall, these observations suggest that transformer-based architectures are particularly well-suited for EEG-based tasks in the context of AD detection, outperforming traditional and earlier deep learning approaches in both speed and stability of convergence.

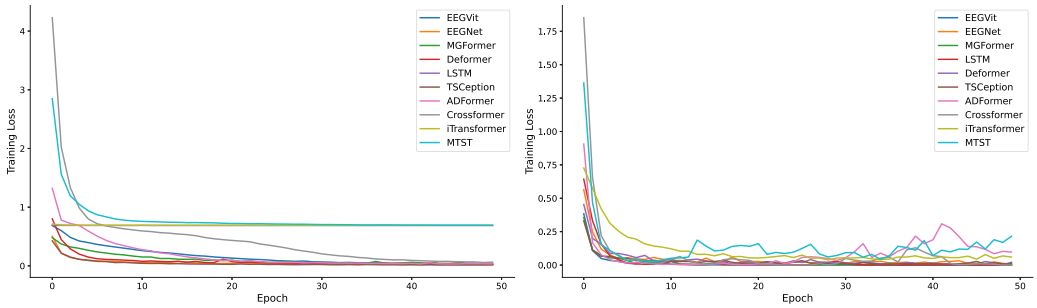
### 5.2. Test results

**AUT-AD.** We evaluated the performance of transformer-based and CNN-based methods using the AUT-AD EEG dataset under a subject-independent setup, ensuring the training and testing sets were strictly separated based on subject IDs. As summarized in Table 2, our proposed MGFormer method significantly outperforms all baseline approaches across multiple evaluation metrics. Specifically, MGFormer achieves an accuracy of 70.48% ± 0.64, precision of 71.80% ± 1.14, recall of 70.48% ± 1.08, and the highest F1-score of 70.50% ± 1.08. Notably, MGFormer also exhibits robust performance in terms of AUROC (83.20% ± 0.91) and AUPRC (83.66% ± 0.92), indicating its strong discriminative capability. In contrast, baseline methods demonstrate lower overall performances. The strongest baseline in terms of accuracy is Crossformer, achieving 62.35% ± 9.38; however, it exhibits high variance, limiting its reliability. EEGViT achieves slightly better consistency, particularly in precision (60.70% ± 3.48) and AUPRC (73.20% ± 3.00), suggesting that ViT-based structures capture EEG characteristics better than simpler Transformer variants such as iTransformer, which significantly underperforms, reporting notably low precision (30.41% ± 0.00) and F1-score (39.20% ± 0.00).

**FSU-AD.** Similarly, we evaluated model performances on the FSU-AD EEG dataset using the same subject-independent evaluation protocol. As presented in Table 2, our proposed MGFormer method again outperforms all baseline methods, achieving exceptional performance across all metrics. MGFormer reports the highest accuracy (97.85% ± 0.65), precision (97.88% ± 0.67), recall (97.85% ± 0.65), and F1-score (97.84% ± 0.65). Furthermore, MGFormer attains remarkable AUROC (99.74% ± 0.19) and AUPRC (99.55% ± 0.32), underscoring its ability to accurately distinguish AD cases within EEG data under subject-independent conditions. Among the baselines, ADFormer and TSCeption deliver competitive results, with TSCeption achieving slightly better accuracy (96.84% ± 0.90) and precision (96.86% ± 0.93) compared to other baselines. ADFormer similarly achieves impressive performance, notably in AUROC (99.67% ± 0.50) and AUPRC (99.49% ± 0.71). However, other methods such as Crossformer and EEGNet demonstrate relatively lower performance, particularly Crossformer, which achieves the lowest accuracy (87.22% ± 2.17) among evaluated models. Overall, MGFormer's substantial improvements over the state-of-the-art confirm its effectiveness in exploiting underlying EEG biomarkers of AD, showing outstanding generalization capability across different subject populations.

### 5.3. Receiver Operating Characteristic (ROC) curve

The Receiver Operating Characteristic (ROC) curves provide valuable insights into model performance by illustrating the balance between true positive and false positive rates. According to Fig. 5(a), on the FSU-AD dataset, all models demonstrated strong classification capabilities, with MGFormer achieving the highest Area Under the Curve (AUC) at 0.997-indicating near-perfect discrimination between Alzheimer's and non-Alzheimer's EEG patterns. While ADFormer (AUC = 0.996) and TSCeption (AUC = 0.995) followed closely, MGFormer maintained a clear edge. In contrast, the AUT-AD dataset in Fig. 5(b)

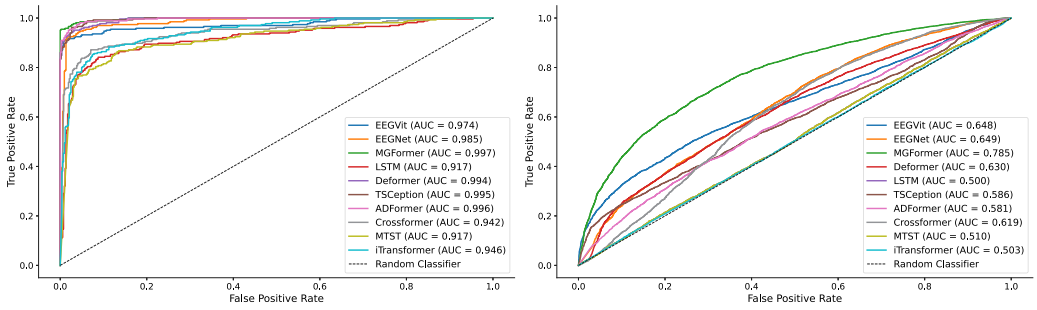



(a) AUT-AD Loss (b) FSU-AD Loss

**Fig. 4. Training loss curves over 50 epochs (for each seed) for SOTA models on AUT-AD and FSU-AD datasets.** MGFormer and ADFormer exhibit the fastest convergence and lowest final loss, indicating efficient learning. Traditional models like EEGNet converge slower and to higher loss values.

**Table 2**
**Results of Subject-Independent Setup.** The training and test sets are distributed based on subjects according to a predetermined ratio/IDs. Results of the FSU-AD and AUT-AD datasets under this setup are presented here. The highest values are denoted in boldface, and the second highest are underlined.

| Datasets | Models | Accuracy | Precision | Recall | F1 score | AUROC | AUPRC |
|---|---|---|---|---|---|---|---|
| AUT-AD (66 participants) | iTransformer | 55.15 ±0.00 | 30.41 ±0.00 | 55.15 ±0.00 | 39.20 ±0.00 | **50.63 ±0.89** | 55.81 ±0.55 |
| | ADFormer | 56.93 ±1.50 | 57.45 ±1.69 | 56.93 ±1.50 | 57.02 ±1.49 | 59.56 ±2.86 | 65.19 ±3.45 |
| | Crossformer | 62.35 ±9.38 | 62.26 ±9.87 | 62.35 ±9.38 | 62.35 ±9.38 | 65.03 ±12.02 | 65.03 ±8.16 |
| | MTST | 53.25 ±0.80 | 51.41 ±0.79 | 53.25 ±0.80 | 49.21 ±2.58 | 50.98 ±1.06 | 56.17 ±0.90 |
| | LSTM | 55.15 ±0.00 | 30.41 ±0.00 | 55.15 ±0.00 | 39.20 ±0.00 | 50.00 ±0.00 | 55.15 ±0.00 |
| | Deformer | 59.13 ±2.43 | 60.27 ±2.72 | 59.13 ±2.43 | 59.15 ±2.43 | 63.20 ±0.04 | 67.40 ±0.03 |
| | EEGViT | 60.09 ±3.18 | 60.70 ±3.48 | 60.09 ±3.18 | 60.05 ±3.21 | 64.60 ±4.00 | 73.20 ±3.00 |
| | EEGNet | 57.99 ±1.93 | 59.62 ±2.03 | 57.99 ±1.93 | 57.81 ±2.04 | 64.00 ±2.00 | 67.00 ±1.00 |
| | TSCeption | 56.21 ±1.54 | 57.99 ±1.35 | 56.21 ±1.54 | 55.94 ±1.72 | 58.40 ±6.60 | 67.00 ±7.60 |
| | MGFormer (ours) | **70.48 ±1.08** | **71.80 ±1.14** | **70.48 ±1.08** | **70.50 ±1.08** | **78.51 ±1.46** | **81.48 ±1.37** |
| FSU-AD(48 participants) | iTransformer | 88.73±1.47 | 88.85 ±1.38 | 88.73 ±1.47 | 88.71±1.44 | 95.06±0.73 | 92.37±1.52 |
| | ADFormer | 96.20 ±1.39 | 96.45 ±1.26 | 96.20 ±1.39 | 96.18 ±1.39 | 99.67 ±0.50 | 99.49 ±0.71 |
| | Crossformer | 87.22 ±2.17 | 88.20 ±2.07 | 87.22 ±2.17 | 87.42 ±2.12 | 94.70 ±1.53 | 92.20 ±2.41 |
| | MTST | 86.71 ±1.50 | 87.09 ±1.11 | 86.71 ±1.50 | 86.76 ±1.38 | 92.24 ±1.07 | 87.42 ±1.56 |
| | LSTM | 87.72 ±1.63 | 88.16 ±1.28 | 87.72 ±1.63 | 87.80 ±1.56 | 93.06 ±1.52 | 90.45 ±2.68 |
| | Deformer | 96.20 ±1.20 | 96.21 ±1.21 | 96.20 ±1.20 | 96.20 ±1.20 | 99.00 ±0.62 | 98.54 ±0.79 |
| | EEGViT | 96.08 ±0.74 | 96.15 ±0.74 | 96.08 ±0.74 | 96.04 ±0.75 | 97.14 ±2.05 | 96.98 ±1.58 |
| | EEGNet | 94.30 ±3.42 | 94.76 ±2.85 | 94.30 ±3.42 | 94.37 ±3.33 | 98.45 ±1.06 | 97.06 ±2.37 |
| | TSCeption | 96.84 ±0.90 | 96.86 ±0.93 | 96.84 ±0.90 | 96.83 ±0.90 | 99.63 ±0.14 | 99.35 ±0.22 |
| | MGFormer (ours) | **97.85 ±0.65** | **97.88 ±0.67** | **97.85 ±0.65** | **97.84 ±0.65** | **99.74 ±0.19** | **99.55 ±0.32** |




(a) FSU-AD ROC Curve (b) AUT-AD ROC Curve

**Fig. 5. Receiver Operating Characteristic (ROC) Curves** for (a) FSU-AD and (b) AUT-AD datasets. The ROC curves illustrate each model's ability to distinguish between AD patients and healthy controls. On the FSU-AD dataset, MGFormer achieves the highest AUC (0.997), indicating near-perfect classification performance, followed closely by ADFormer and TSCeption.

posed a greater challenge, yielding generally lower AUC values; yet, MGFormer again outperformed all other models with an AUC of 0.785. These results shows MGFormer's superior generalizability and robustness across varying EEG datasets, especially under more complex or noisy conditions.

### 5.4. Ablation study

In this study we conduct several ablation analysis, including effects of number of HFF layers, different feature extraction methods from these layers and kernel sizes.

#### 5.4.1. Ablation study of FFT-features and 1D CNN

We tested our models performance by removing parts of the model, namely FFT-feature extractors and 1-D CNN inside HFF. Table 3 presents the ablation results, where removing the 1D-CNN encoder from within the HFF module led to a notable decline in performance, particularly on the AUT-AD dataset, where accuracy dropped significantly from 70.48% to 60.57%, indicating that the CNN encoder substantially contributes to the model's ability to capture essential temporal features. On the other hand, the exclusion of FFT features resulted in only slight decreases in performance, with accuracy on AUT-AD modestly decreasing from 70.50% to 68.98% and similarly minor reductions

**Table 3**
**Results of ablation without FFT features and 1D CNN encoder within the HFF.** Removal of 1D-CNN had drastic effect on AUT-AD, while FFT-feature removal has a milder one.

| Datasets | Kernels | Accuracy | Precision | Recall | F1 score | AUROC | AUPRC |
|---|---|---|---|---|---|---|---|
| AUT-AD (66 participants) | w/o FFT Features | 68.98 ±1.60 | 70.00 ±1.38 | 68.98 ±1.60 | 68.99 ±1.64 | 76.81 ±2.02 | 80.29 ±1.46 |
| | w/o CNN inside HFF | 60.57 ±3.12 | 62.36 ±2.95 | 60.57 ±3.12 | 60.39 ±3.29 | 66.97 ±4.07 | 68.75 ±2.94 |
| | MGFormer (ours) | **70.48 ±1.08** | **71.80 ±1.14** | **70.48 ±1.08** | **70.50 ±1.08** | **78.51 ±1.46** | **81.48 ±1.37** |
| FSU-AD (48 participants) | w/o FFT Features | **98.23 ±0.74** | **98.24 ±0.75** | **98.23 ±0.74** | **98.23 ±0.74** | 99.58 ±0.41 | 99.40 ±0.48 |
| | w/o CNN inside HFF | 97.97 ±1.29 | 98.01 ±1.27 | 97.97 ±1.29 | 97.97 ±1.29 | **99.69 ±0.35** | **99.55 ±0.43** |
| | MGFormer (ours) | 97.85 ±0.65 | 97.86 ±0.63 | 97.85 ±0.65 | 97.84 ±0.65 | 99.68 ±0.21 | 99.46 ±0.33 |

**Table 4**
**Results of layer-wise ablation analysis.** Layer-wise ablation study examining the impact of HFF layer depth on model performance. We observed a performance plateau beyond a threshold, indicating reduced information content in deeper embeddings, which is supported by the results. The highest values are denoted in boldface, and the second highest are underlined.

| Datasets | Layers | Accuracy | Precision | Recall | F1 score | AUROC | AUPRC |
|---|---|---|---|---|---|---|---|
| AUT-AD(66 participants) | 1 | 67.83 ±1.60 | 70.93 ±1.69 | 67.83 ±1.60 | 67.51 ±1.68 | 76.94 ±1.85 | 79.55 ±2.62 |
| | 2 | 69.32 ±1.47 | 71.03 ±1.29 | 69.32 ±1.47 | 69.28 ±1.52 | 77.43 ±1.67 | 80.13 ±1.99 |
| | 4 | **70.48 ±1.08** | **71.80 ±1.14** | **70.48 ±1.08** | **70.50 ±1.08** | **78.51 ±1.46** | **81.48 ±1.37** |
| | 6 | 68.94 ±1.02 | 70.33 ±1.53 | 68.94 ±1.02 | 68.93 ±0.98 | 76.39 ±0.73 | 79.06 ±1.40 |
| | 8 | 68.31 ±1.49 | 69.92 ±0.89 | 68.31 ±1.49 | 68.26 ±1.56 | 76.07 ±1.83 | 79.19 ±1.73 |
| FSU-AD(48 participants) | 1 | **97.85 ±0.65** | **97.86 ±0.63** | **97.85 ±0.65** | **97.84 ±0.65** | 99.68 **±0.21** | 99.46 ±0.33 |
| | 2 | 97.47 ±0.90 | 97.47 ±0.89 | 97.47 ±0.90 | 97.46 ±0.90 | 99.67 ±0.25 | 99.45 ±0.39 |
| | 4 | 97.47 ±1.74 | 97.50 ±1.73 | 97.47 ±1.74 | 97.45 ±1.77 | **99.71 ±0.30** | **99.49 ±0.51** |
| | 6 | 96.84 ±1.06 | 96.84 ±1.07 | 96.84 ±1.06 | 96.83 ±1.06 | 99.50 ±0.24 | 99.20 ±0.33 |
| | 8 | 97.72 ±0.95 | 97.73 ±0.94 | 97.72 ±0.95 | 97.71 ±0.95 | 99.60 ±0.63 | 99.49 ±0.71 |

**Table 5**
**Results of kernel-wise ablation analysis.** We study the effects of the combinations of temporal kernels on model performance. Kernels 1, 2, 3 refers to the set of kernels in $f \times \mathcal{G}$; where $\mathcal{G} = \{0.04, 0.06, 0.08\}$. The highest values are denoted in boldface, and the second highest are underlined.

| Datasets | Kernels | Accuracy | Precision | Recall | F1 score | AUROC | AUPRC |
|---|---|---|---|---|---|---|---|
| AUT-AD(66 participants) | {1} | 67.71 ±1.82 | 69.63 ±1.55 | 67.71 ±1.82 | 67.61 ±1.91 | 74.83 ±2.49 | 80.05 ±1.39 |
| | {2} | 68.08 ±2.89 | 69.68 ±2.43 | 68.08 ±2.89 | 67.99 ±3.03 | 75.04 ±3.17 | 77.66 ±2.56 |
| | {3} | 69.35 ±1.49 | 70.04 ±1.16 | 69.35 ±1.49 | 69.35 ±1.46 | 75.82 ±0.92 | 76.44 ±1.07 |
| | {1, 2} | 68.70 ±1.17 | 70.42 ±2.16 | 68.70 ±1.17 | 68.62 ±1.09 | 76.69 ±2.41 | 80.69 ±2.25 |
| | {2, 3} | **70.85 ±1.41** | **71.93 ±0.88** | **70.85 ±1.41** | **70.86 ±1.50** | 78.45 ±0.73 | 79.91 ±1.14 |
| | {1, 2, 3} | 70.48 ±1.08 | 71.80 ±1.14 | 70.48 ±1.08 | 70.50 ±1.08 | **78.51 ±1.46** | **81.48 ±1.37** |
| FSU-AD(48 participants) | {1} | 97.47 ±0.90 | 97.53 ±0.88 | 97.47 ±0.90 | 97.47 ±0.89 | **99.81 ±0.09** | **99.66 ±0.15** |
| | {2} | 95.95 ±2.76 | 96.13 ±2.61 | 95.95 ±2.76 | 95.95 ±2.76 | 99.34 ±0.62 | 98.78 ±1.19 |
| | {3} | 96.20 ±1.06 | 96.24 ±1.01 | 96.20 ±1.06 | 96.21 ±1.05 | 99.32 ±0.38 | 98.94 ±0.53 |
| | {1, 2} | 97.34 ±1.09 | 97.35 ±1.09 | 97.34 ±1.09 | 97.33 ±1.09 | 99.53 ±0.30 | 99.25 ±0.44 |
| | {2, 3} | 96.20 ±1.65 | 96.38 ±1.46 | 96.20 ±1.65 | 96.22 ±1.61 | 99.26 ±0.54 | 98.73 ±0.95 |
| | {1, 2, 3} | **97.72 ±0.95** | **97.73 ±0.94** | **97.72 ±0.95** | **97.71 ±0.95** | 99.60 ±0.63 | 99.49 ±0.71 |

observed across other metrics. This suggests that, although FFT features enhance performance, the 1D-CNN module plays a more critical role in capturing discriminative patterns necessary for robust classification, particularly for more complex datasets like AUT-AD. Interestingly, on the FSU-AD dataset, the impact of both FFT feature removal and 1D-CNN encoder removal was less pronounced, indicating dataset-specific sensitivities and underscoring the importance of the 1D-CNN encoder predominantly in contexts with higher temporal sequence. Since AUT-AD was segmented into 5 s epochs compared to FSU-AD's 1 s epoch, we believe both FFT-features and 1-D CNN has much profound impact on longer sequence of data.

### 5.4.2. Effects of changing HFF layer count

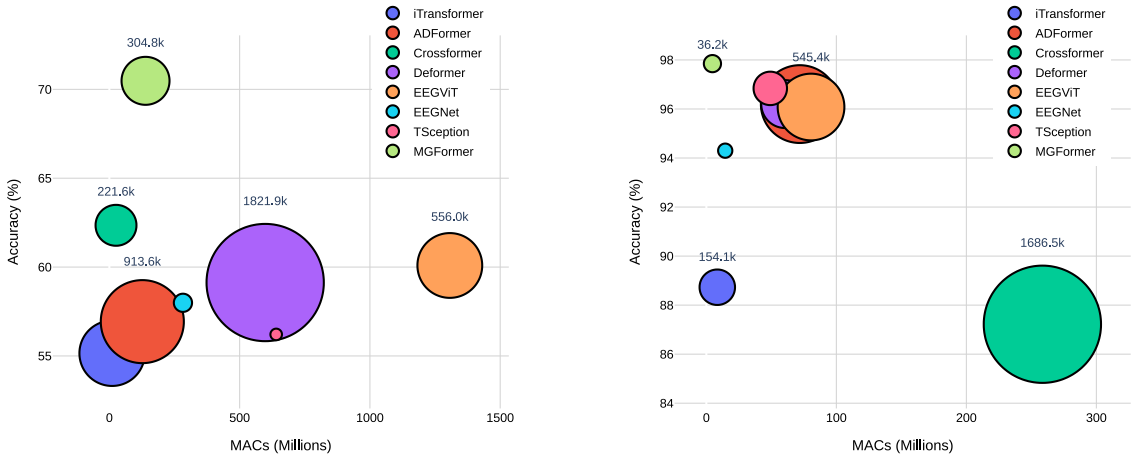
In our layer-wise ablation analysis in Table 4, we observed that increasing the HFF depth beyond a certain point does not necessarily translate to better performance, supporting our hypothesis that deeper embeddings can reduce informative content. For the AUT-AD dataset, performance peaks at four layers with an accuracy of 70.48% and F1 score of 70.50%, after which adding more layers yields diminishing returns. Notably, for the FSU-AD dataset, a single-layer configuration already achieves state-of-the-art results (97.85% accuracy and 97.84% F1 score), indicating that an overly deep model may dilute rather than enhance representation quality. These findings underscore the importance of carefully selecting the appropriate network depth for different datasets, rather than assuming "more is better".

### 5.4.3. Different kernel sizes and it is effects

The kernel-wise ablation analysis in Table 5 reveals that different kernel configurations can substantially impact classification performance across various metrics and datasets. On the AUT-AD dataset, using kernels 2, 3 yields the highest accuracy, precision, recall, and F1 scores, while including all three kernels (1, 2, 3) gives the best AUROC and AUPRC. This suggests that although kernels 2, 3 capture critical temporal features that translate directly into better point-estimate metrics (accuracy, precision, recall, F1), the additional kernel 1 helps broaden the temporal receptive field, leading to slightly improved area-under-curve measures (AUROC, AUPRC). In contrast, on the FSU-AD dataset, the single kernel 1 achieves the highest AUROC and AUPRC but is slightly outperformed by the full triple-kernel combination (1, 2, 3) on the standard classification metrics. These findings underscore the importance of multi-scale temporal modeling: certain kernels better capture short-range dependencies that boost precision and recall, whereas broader or finer temporal kernels may enhance separability as reflected in area-under-curve metrics. Consequently, the choice of kernel combination should be guided not only by raw classification accuracy but also by which performance measures are most critical for the application at hand.

## 5.5. Parameter comparison with other baseline models

We also compare our model's parameter efficiency against other baseline models. Specifically, we use three commonly referenced metrics-Floating-Point Operations Per Second (FLOPS), Multiply-Accumulate Operations (MACs), and parameter count-to assess computational




(a) Parameter comparison for AUT-AD

(b) Parameter comparison for FSU-AD

**Fig. 6. Comparison of model performance on both datasets in terms of accuracy vs. computational complexity (MACs in thousands).** Bubble size represents the number of parameters in each model. MGFormer achieves the highest accuracy with the lowest MACs and smallest parameter count across two datasets.

**Table 6**
**Paired t-test results comparing MGFormer with baseline models on two EEG datasets.** The table reports p-values from paired t-tests and indicates whether the performance difference between MGFormer and each baseline model is statistically significant ($p < 0.05$). Results are presented separately for the AUT-AD and FSU-AD datasets. Statistically significant differences are highlighted in bold.

| Models | AUT-AD | | FSU-AD | |
|---|---|---|---|---|
| | P-value | Significance | P-value | Significance |
| Deformer | 0.003 | **Yes** | 0.002 | **Yes** |
| EEGNet | 0.000 | **Yes** | 0.000 | **Yes** |
| TSCeption | 0.001 | **Yes** | 0.195 | No |
| EEGViT | 0.001 | **Yes** | 0.013 | **Yes** |
| iTrasnformer | 0.000 | **Yes** | 0.000 | **Yes** |
| Crossformer | 0.087 | No | 0.000 | **Yes** |
| ADFormer | 0.000 | **Yes** | 0.165 | No |
| MTST | 0.000 | **Yes** | 0.000 | **Yes** |
| LSTM | 0.000 | **Yes** | 0.000 | **Yes** |

performance and capacity [14]. FLOPS measures the rate at which a system can perform floating-point calculations, providing a sense of the hardware's raw computational capability. Meanwhile, MACs capture the multiply-and-accumulate operations fundamental to deep learning workloads, making this metric a more direct reflection of the computational demands tied to the model architecture. Lastly, parameter count indicates the number of trainable variables (weights, biases, or other parameters) in a model, signifying its capacity to learn complex patterns; however, a higher parameter count typically requires more resources and larger datasets to avoid overfitting. As shown in Fig. 6, MGFormer demonstrates superior performance over the baselines by achieving fewer FLOPs and MACs, all while maintaining higher accuracy.

### 5.6. Statistical significance

To confirm the significance of MGFormer's improvements, paired t-tests compared its performance against nine baseline models on two EEG datasets (AUT-AD and FSU-AD). Table 6 summarizes the results. On the AUT-AD dataset, MGFormer significantly outperformed eight of nine baselines, including EEGNet, iTransformer, and MTST (all $p < 0.001$), but not Crossformer. On the FSU-AD dataset, MGFormer showed significant improvements over seven of nine baselines, with non-significant differences compared to TSCeption (p=0.20) and ADFormer (p=0.17). These results confirm MGFormer's consistent and reliable advantage across the models and datasets.

## 6. Model interpretation

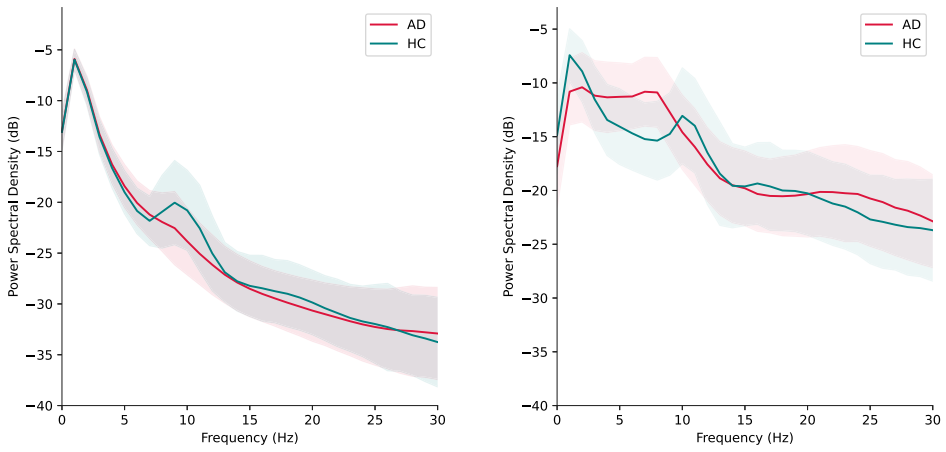
### 6.1. Power Spectral Density (PSD)

Power Spectral Density (PSD) describes how the power of a signal is distributed over different frequencies [48]. Essentially, it decomposes a signal from the time domain into its frequency components, showing which frequencies contain the most "power" or energy per unit bandwidth. This allows for the identification of dominant frequencies and the characterization of noise or other frequency dependent phenomena within a signal. The PSD $P_{xx}(f)$ is given by:

$$P_{xx}(f) = \frac{1}{LU} \sum_{l=0}^{L-1} \left| \sum_{n=0}^{N-1} w(n) x_l(n) e^{-j2\pi f n} \right|^2 \tag{13}$$

where $x_l(n)$ is the $l$th segment of the signal, $w(n)$ is a window function, $L$ is the number of segments, $N$ is the length of each segment, and $U$ is a normalization factor accounting for the power of the window.

Fig. 7 shows PSD for AUT-AD and FSU-AD respectively. Studies indicate that AD patients exhibit "slowing of the EEG" compared to healthy individuals. This means an increase in delta (0.5–4 Hz) power, alongside a decrease in alpha band (8–13 Hz) [49]. We can observe this in both datasets. However, it is important to mention that FSU-AD is an eyes-open dataset, which naturally caused a relative suppression in alpha band [50], compared to AUT-AD which is eyes-closed.




(a) AUT-AD (b) FSU-AD

**Fig. 7. Power Spectral Density (PSD)** for (a) AUT-AD and (b) FSU-AD datasets. The green curve represents HC, and the red curve represents AD patients, with shaded regions or error bars indicating variability. In both datasets, AD patients exhibit a noticeable reduction in alpha band (8–13 Hz) power, consistent with EEG slowing. The AUT-AD dataset (eyes-closed) shows clearer alpha activity than the FSU-AD dataset (eyes-open), where alpha suppression is expected. The smoother PSD curve in (a) is due to a larger dataset and averaging across more segments.

### 6.2. Functional brain connectivity

To identify connectivity patterns, we analyzed EEG recordings using the Phase Lag Index (PLI), which captures functional synchrony. EEG signals were bandpass filtered in the alpha range (8–13 Hz), a frequency band sensitive to cognitive decline [49], using a 4th-order Butterworth filter. Instantaneous phases were extracted using the Hilbert transform, and PLI was computed between all channel pairs to form a symmetric connectivity matrix. The top 30% strongest connections were visualized using a graph layout based on the standard 10–20 EEG system. Nodes were scaled by degree, and edges were color-coded: blue (left hemisphere), red (right hemisphere), and teal (inter-hemispheric). In the top row, connectivity graphs generated using PLI reveal marked differences between AD and HC groups. Specifically, AD brains (Figures a and c) exhibit disrupted connectivity characterized by reduced inter-hemispheric links, fewer long-range connections, and more spatially clustered or asymmetric patterns. These features suggest a breakdown of global integration and a shift toward localized activity, which is consistent with known disruptions in brain network organization in Alzheimer's disease.

In contrast, control groups (Fig. 8(b) and (d)) shows more symmetrical and widespread connectivity, indicating preserved global communication pathways. The lower panels (Fig. 8(e) - (h)) present the corresponding PLI heatmaps, capturing the strength of phase synchronization across all electrode pairs. AD subjects show pockets of increased PLI values, particularly in posterior and central areas (Fig. 8(e) and (g)), which may reflect compensatory overconnectivity or regional network imbalance. Meanwhile, control subjects show more homogeneous and lower PLI distributions (f, h), consistent with stable and efficient neural communication [10] Together, these findings highlight the diagnostic potential of PLI-based functional connectivity in identifying AD-related neural alterations.

### 6.3. Saliency map

We applied Grad-CAM [51] to visualize our model's saliency maps across brain topography for both datasets. Notably, we observed increased model attention in the temporal (T4/T5), parietal (P3/P4), and central (Cz/C3/C4) regions among AD patients, as shown in Fig. 9. This observation aligns with existing research, including temporal lobe atrophy highlighted in [52], parietal region involvement identified in [53], and central region alterations described in [54], each suggesting distinctive biomarker characteristics associated with AD. Therefore, our model's saliency maps clearly correspond with established AD biomarkers. Interestingly, the saliency maps for healthy controls (HC) exhibited notably different patterns, particularly for the FSU-AD dataset, which involves eye-open tasks, prompting the model to emphasize the occipital region.

## 7. Discussion

This study introduced MGFormer, a lightweight CNN-Transformer hybrid architecture, and validated its efficacy for subject-independent AD classification from EEG signals. We evaluated our model comprehensively on 2 complete datasets, totaling 113 subjects and 11,383 segments. The model achieved accuracies of 70.48% on the AUT-AD dataset and 97.85% on the FSU-AD dataset, outperforming nine state-of-the-art baseline models across six evaluation metrics. To ensure our model's effectiveness in real-world situations, we tested it using a strict evaluation protocol that ensures the data used to train and test it come from different individuals. This approach gives us a clearer picture of how well our model can adapt to different patients and clinical environments.

The success of MGFormer is attributable to two primary architecture designed to address the unique properties of medical time series data. First, the MgTE module extracts spatio-temporal features at multiple granularities by using temporal kernels of varying lengths. This multi-scale processing allows the model to simultaneously capture both high-frequency local fluctuations and broader, long-range contextual information within the signal. Second, the HFF module integrates three distinct processing streams: a self-attention mechanism to model long-range temporal dependencies, a 1D CNN to capture local features [14], and an FFT-based extractor for spectral information. Ablation studies confirmed that both the 1D CNN and FFT-based features were critical contributors to the model's overall performance. MGFormer accomplishes this high accuracy while maintaining superior computational efficiency. Model complexity analysis show that MGFormer has a lower parameter count and fewer MAC than competing models, addressing a common limitation of computationally intensive Transformer architectures. This lightweight design is essential for practical deployment in clinical environments where computational resources may be limited.

### 7.1. Future work

The promising results demonstrated by MGFormer suggest several potential avenues for future exploration and improvement. Firstly,

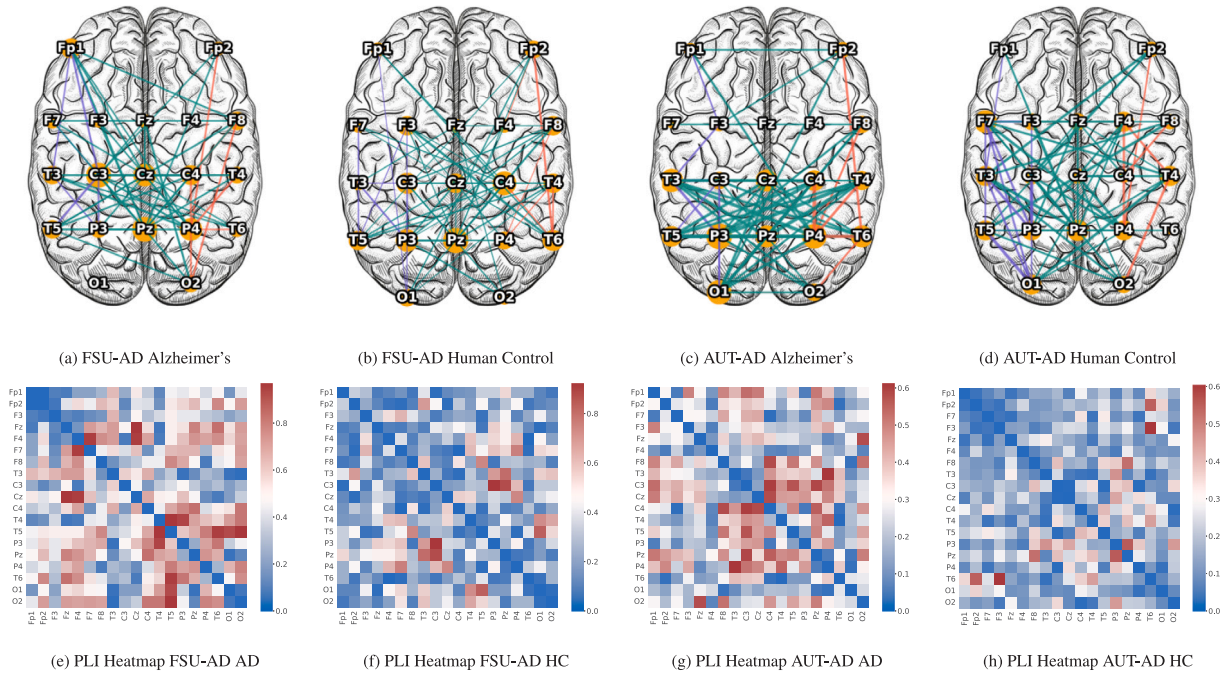
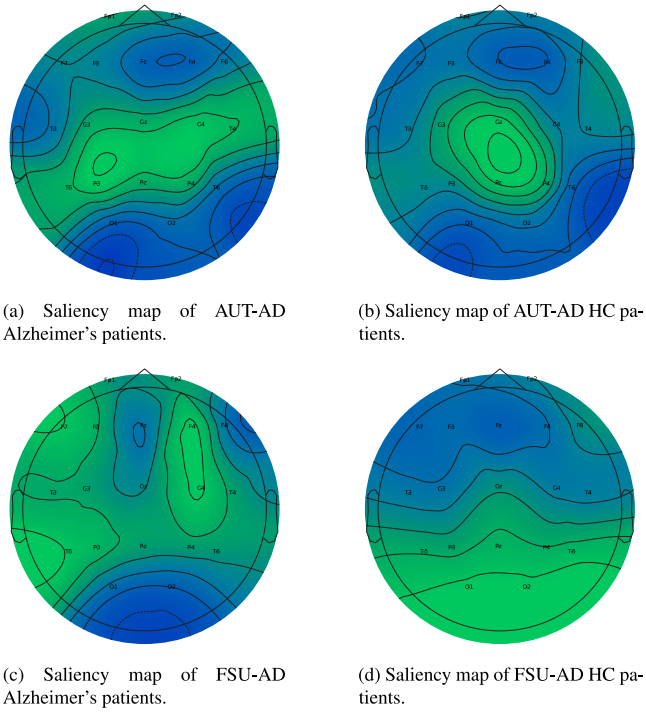



**Fig. 8. PLI-based functional connectivity and heatmaps for AD and control groups.** EEG signals were bandpass filtered in the alpha band (8–13 Hz), and functional connectivity was computed using the Phase Lag Index (PLI). Top-row graphs display the top 30% strongest connections, revealing reduced inter-hemispheric and long-range connectivity in AD subjects (a, c), compared to the more symmetric and distributed networks in controls (b, d). Bottom-row heatmaps (e–h) show increased local synchrony in AD, particularly in posterior regions, while controls exhibit more homogeneous connectivity.



(a) Saliency map of AUT-AD Alzheimer's patients.

(b) Saliency map of AUT-AD HC patients.

(c) Saliency map of FSU-AD Alzheimer's patients.

(d) Saliency map of FSU-AD HC patients.

**Fig. 9. Saliency maps for AD patients and healthy controls (HC) across both datasets reveal distinct patterns.** In AD patients, our model consistently focused on the temporal (T4/T5), parietal (P3/P4), and central (Cz/C3/C4) regions-aligning with findings from [55]. Interestingly, we observed increased saliency in occipital channels for HC subjects in the FSU-AD dataset, since FSU-AD had eye open tasks as well. Which suggests heightened visual processing activity in that group.

investigating alternative transformer architectures, such as sparse attention mechanisms or efficient transformers, may further enhance computational efficiency and model scalability, reducing training time in resource-limited clinical settings. We have used fixed positional embedding and kernels, where future users can replace it with learnable ones. Moreover, the current study has focused primarily on AD versus HC classification. Future studies could extend MGFormer to multi-class scenarios involving mild cognitive impairment (MCI) and other types of dementia, such as Frontotemporal Dementia (FTD), to evaluate its efficacy in broader diagnostic contexts. Additionally, exploring the application of explainability techniques like attention-map visualization could provide deeper insights into the specific EEG features driving the model's predictions, increasing clinical trust and interpretability. Similarly, integrating complementary data modalities-such as structural and functional magnetic resonance imaging (MRI), genetic markers, and clinical scores-with EEG signals may improve diagnostic accuracy and provide better patient assessments. Lastly, validating MGFormer in prospective longitudinal studies would be crucial for assessing its potential utility in monitoring disease progression and evaluating treatment efficacy in clinical trials.

### 7.2. Limitations

While MGFormer has shown strong performance in EEG-based AD classification, several limitations should be acknowledged. First, the evaluation was conducted on only two publicly available datasets (AUT-AD and FSU-AD), which may limit the generalizability of the findings to other populations or EEG collection settings. Future work should consider validation on more diverse and independent cohorts. Second, despite measures taken to prevent overfitting, the model may still learn dataset-specific EEG characteristics, particularly given the relatively small sample sizes in clinical EEG datasets. Finally, the current model focuses solely on binary classification (AD vs. HC), and its performance in more clinically realistic, multi-class scenarios remains to be established. Addressing these limitations will be essential for the reliable deployment of MGFormer in real-world clinical applications.

## 8. Conclusion

This study tackled the important challenge of accurately identifying Alzheimer's disease (AD) through EEG signals, especially in realistic situations where the testing subjects differ from those seen during training. Previously, most EEG-based methods have depended heavily on convolutional neural networks (CNNs) or manually extracted statistical features. However, these approaches often overlook the potential benefits of Transformers, which excel at capturing global context and long-range temporal relationships in EEG data. Our results indicate that MGFormer, a CNN-Transformer-based model designed to operate across multiple granularities, significantly outperformed other leading CNN and transformer methods on various evaluation metrics. Specifically, MGFormer achieved impressive accuracy rates of 70.48% on the AUT-AD dataset and 97.85% on the FSU-AD dataset, along with higher precision, recall, and F1 scores in the challenging, realistic scenario of subject-independent testing. The success of MGFormer can be attributed to its Multi-Granularity Token Encoding (MgTE) module, which effectively extracts both local details and broader EEG patterns, and its Hybrid Feature Fusion (HFF) module, which combines convolutional processing with frequency-domain information. The implications of our findings are highly significant for both clinical practice and future research. Additionally, detailed ablation studies provided valuable insights into optimizing kernel sizes, transformer layers, and feature extraction techniques, helping guide future EEG-based diagnostic tools. Ultimately, this research underscores the necessity of creating EEG models that can generalize across diverse individuals, opening exciting opportunities for personalized and widely applicable neurological assessments.

## CRediT authorship contribution statement

**Raiyan Rahman:** Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Abul Kalam al Azad:** Writing – original draft, Validation, Supervision, Project administration, Investigation, Funding acquisition. **Sifat Momen:** Writing – original draft, Validation, Supervision, Project administration, Funding acquisition, Conceptualization.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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## Appendix. Models

### A.1. iTransformer

The iTransformer [26] is a novel time series forecasting model that repurposes the Transformer architecture by inverting its input dimensions; treating each time series variate as a token instead of each timestamp. Unlike traditional Transformers, which apply attention over time, iTransformer applies self-attention across variates to capture multivariate correlations and uses feed-forward networks to model temporal dynamics. This inversion avoids issues from combining different variates at the same timestamp (e.g., misaligned or heterogeneous data) and enables better scalability and interpretability. Experiments show iTransformer outperforms state-of-the-art models on a wide range of datasets, especially in high-dimensional and long-lookback scenarios. Importantly, it does this without modifying core Transformer components, making it compatible with many efficient attention mechanisms.

### A.2. ADformer

The ADformer [13] architecture integrates two parallel branches to extract complementary temporal and spatial features at multiple granularities. The patch branch explicitly captures temporal features through cross-channel multi-granularity patch embedding and self-attention while implicitly capturing spatial correlations. Conversely, the channel branch explicitly models spatial correlations through whole-series multi-granularity channel embedding and self-attention, implicitly encoding temporal information. Both branches employ data augmentation techniques to encourage the learning of mutual information and feature complementarity among granularities, enhanced by intra- and inter-granularity self-attention mechanisms facilitated by routers for computational efficiency. Finally, the combined embeddings from both branches form a robust representation for effective downstream classification tasks.

### A.3. Crossformer

The Crossformer model [41] addresses limitations of existing Transformer-based models in multivariate time series (MTS) forecasting by explicitly capturing both cross-time and cross-dimension dependencies. It introduces a Dimension-Segment-Wise (DSW) embedding technique that converts time series data into a 2D vector representation, preserving distinct temporal and dimensional information. The core innovation is the Two-Stage Attention (TSA) layer, which efficiently captures cross-time dependencies within dimensions and cross-dimension dependencies across variables through a router mechanism. Crossformer utilizes these components within a Hierarchical Encoder-Decoder (HED), capturing dependencies at multiple scales by merging segments at progressively coarser resolutions. Experimental evaluations across six real-world datasets demonstrate that Crossformer significantly outperforms previous state-of-the-art models.

### A.4. Deformer

EEG-Deformer [14] is a Transformer-based model designed for decoding EEG signals in brain-computer interfaces (BCI), consisting of three components: a shallow CNN-based feature encoder, a Hierarchical Coarse-to-fine Transformer (HCT), and a Dense Information Purification (DIP) module. The shallow encoder initially extracts temporal and spatial features using two-layer CNNs, which are then rearranged and combined with positional encodings for input into the Transformer. Next, the HCT captures both coarse-grained (long-term) and fine-grained (short-term) temporal dynamics using parallel Transformer-based and CNN-based branches, integrating these via sum fusion. DIP further refines and progressively fuses the fine-grained features from multiple HCT blocks, using power features that represent EEG frequencies effectively. Finally, these purified features are concatenated and passed through a linear classifier to produce the decoding output.

### A.5. MTST

The Multiresolution Time-Series Transformer (MTST) is a framework that introduces multi-scale analysis into patch-based time-series transformers for forecasting tasks [42]. Unlike previous methods that employ upsampling or downsampling strategies, MTST utilizes multiple patch-based tokenizations with varying patch sizes, combined with a multi-branch transformer architecture within each layer. This approach enables the model to flexibly capture temporal patterns across different resolutions. Additionally, the integration of relative positional encoding aligns well with the seasonal structures commonly observed in real-world datasets.

### A.6. LSTM

Long-Short Term Memory is a recurrent neural network architecture designed to model long-term dependencies in sequential data [44]. In this work we have used vanilla LSTM which uses a cell state and three gates; input, forget, and output to regulate the flow of information. These gates are sigmoid functions that control what information to retain or discard at each time step.

### A.7. EEGViT

EEGViT used in this work is a version of Vision Transformer [27], adapted by the work of [14]. In EEGViT the segments are transformed into non-overlapping patches, which is subsequently fed into the vision transformer architecture.

### A.8. EEGNet

EEGNet [43] is a compact convolutional neural network composed of two primary convolutional blocks designed specifically for EEG data. The first block applies temporal convolution to extract frequency-based features, followed by depthwise spatial convolutions to learn frequency-specific spatial filters. The second block utilizes separable convolutions, combining depthwise temporal summaries and pointwise convolutions to optimally mix features across maps. Each block incorporates batch normalization, ELU activations, average pooling, and dropout for regularization. The network concludes with a softmax classification layer, providing a lightweight and generalizable architecture effective across various EEG-based BCI tasks.

### A.9. TSCeption

TSception is a multi-scale CNN-based model [29] designed for EEG-based emotion recognition, explicitly capturing temporal dynamics and spatial asymmetry inherent in EEG signals. The model architecture consists of three key layers: a dynamic temporal layer that uses multi-scale 1D convolutional kernels proportional to the EEG sampling rate to learn dynamic time-frequency features; an asymmetric spatial layer employing global and hemisphere-specific kernels to exploit brain hemispheric asymmetry associated with emotions; and a high-level fusion layer, which integrates hemisphere and global features into compact, discriminative representations.

## Data availability

The data are available in Campanharoet al. (2022), Miltiadouset al. (2023). The code is available at https://github.com/Raiyan78/MGformer.

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